Spin along an arbitrary direction

For a spin-$\frac{1}{2}$ eigenstate of S_z, with $s_z = \pm\frac{1}{2}$,

$$\begin{aligned}\langle \tfrac{1}{2}, s_z|\mathbf{S}|\tfrac{1}{2}, s_z\rangle &= \langle \tfrac{1}{2}, s_z|S_x|\tfrac{1}{2}, s_z\rangle\,\hat{\mathbf{e}}_x + \langle \tfrac{1}{2}, s_z|S_y|\tfrac{1}{2}, s_z\rangle\,\hat{\mathbf{e}}_y + \langle \tfrac{1}{2}, s_z|S_z|\tfrac{1}{2}, s_z\rangle\,\hat{\mathbf{e}}_z \\ &= s_z\hbar\hat{\mathbf{e}}_z.\end{aligned}$$

The first two matrix elements vanish. More generally, choose the commuting observables S^2 and $\mathbf{S}\cdot\hat{\mathbf{n}}$. Their simultaneous eigenstates may be denoted by $|\hat{\mathbf{n}}, \pm\rangle$:

$$\begin{aligned}S^2|\hat{\mathbf{n}}, \pm\rangle &= \tfrac{1}{2}(\tfrac{1}{2}+1)\hbar^2|\hat{\mathbf{n}}, \pm\rangle, \\ \mathbf{S}\cdot\hat{\mathbf{n}}|\hat{\mathbf{n}}, \pm\rangle &= \pm\frac{\hbar}{2}|\hat{\mathbf{n}}, \pm\rangle.\end{aligned}$$

Do we obtain

$$\langle \hat{\mathbf{n}}, \pm|\mathbf{S}|\hat{\mathbf{n}}, \pm\rangle = \pm\frac{\hbar}{2}\hat{\mathbf{n}}\,?$$

Let $\hat{\mathbf{n}}$ point in the direction (θ, ϕ):

$$n_z = \cos\theta, \qquad n_x = \sin\theta\cos\phi, \qquad n_y = \sin\theta\sin\phi, \qquad \hat{\mathbf{n}} = n_x\hat{\mathbf{e}}_x + n_y\hat{\mathbf{e}}_y + n_z\hat{\mathbf{e}}_z.$$

In the S_z representation,

$$\begin{aligned}\hat{\mathbf{n}}\cdot\mathbf{S} &= \frac{\hbar}{2}\left[n_x\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} + n_y\begin{pmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{pmatrix} + n_z\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right] \\ &= \frac{\hbar}{2}\begin{pmatrix}n_z & n_x - \mathrm{i}n_y\\ n_x + \mathrm{i}n_y & -n_z\end{pmatrix} \\ &= \frac{\hbar}{2}\begin{pmatrix}\cos\theta & \sin\theta\,\mathrm{e}^{-\mathrm{i}\phi}\\ \sin\theta\,\mathrm{e}^{\mathrm{i}\phi} & -\cos\theta\end{pmatrix}.\end{aligned}$$

Exercise retained from the notes

Find the eigenvalues and eigenvectors of $\hat{\mathbf{n}}\cdot\mathbf{S}$ in the S_z representation. Then calculate the displayed vector expectation value explicitly.

Pauli matrices

In the spin-up/spin-down basis,

$$\mathbf{S} = \frac{\hbar}{2}\boldsymbol{\sigma}, \qquad \{A, B\} \equiv AB + BA.$$

The following identities are useful:

(1) For $i \neq j$, $\{\sigma_i, \sigma_j\} = 0$, or $\sigma_i\sigma_j = -\sigma_j\sigma_i$.
(2) Since $S_xS_y - S_yS_x = \mathrm{i}\hbar S_z$,

$$\sigma_x\sigma_y - \sigma_y\sigma_x = 2\mathrm{i}\sigma_z.$$

Together with (1), this gives $\sigma_x\sigma_y = \mathrm{i}\sigma_z$ and the cyclic permutations.
(3) $\operatorname{Tr}\sigma_i = 0$.
(4) $\sigma_i^2 = I$; more generally, $(\hat{\mathbf{n}}\cdot\boldsymbol{\sigma})^2 = I$ for a unit vector $\hat{\mathbf{n}}$.

Proof of the fourth identity

Write an arbitrary spin-$\frac{1}{2}$ state as

$$|\psi\rangle = \alpha|+\rangle + \beta|-\rangle, \qquad |\pm\rangle \equiv |\hat{\mathbf{n}}, \pm\rangle.$$

Then

$$\left(\hat{\mathbf{n}}\cdot\mathbf{S} + \frac{\hbar}{2}I\right)\left(\hat{\mathbf{n}}\cdot\mathbf{S} - \frac{\hbar}{2}I\right)|\psi\rangle$$
$$= \left(\hat{\mathbf{n}}\cdot\mathbf{S} + \frac{\hbar}{2}I\right)(-\beta\hbar|-\rangle) = 0.$$

Because this holds for every state,

$$(\hat{\mathbf{n}}\cdot\mathbf{S})^2 = \frac{\hbar^2}{4}I, \qquad (\hat{\mathbf{n}}\cdot\boldsymbol{\sigma})^2 = I.$$

Clarification. The intermediate action of the second factor is $-\beta\hbar|-\rangle$; the handwritten intermediate line omits its minus sign. The final identity is unchanged.

Combining these results gives the compact Pauli algebra:

$$\boxed{\{\sigma_i, \sigma_j\} = 2\delta_{ij}I}, \qquad [\sigma_x, \sigma_y] = 2\mathrm{i}\sigma_z \quad \text{and cyclic permutations.}$$

A vector identity for Pauli matrices

The identity

$$\boxed{(\mathbf{A}\cdot\boldsymbol{\sigma})(\mathbf{B}\cdot\boldsymbol{\sigma}) = (\mathbf{A}\cdot\mathbf{B})I + \mathrm{i}(\mathbf{A}\times\mathbf{B})\cdot\boldsymbol{\sigma}}$$

is useful, for example, in the relativistic quantum mechanics of electrons and the Dirac equation. Here the components of $\mathbf{A}$ and $\mathbf{B}$ are numbers, or commute with each other and with the Pauli matrices.

Derivation

Expanding the product gives

$$\begin{aligned}(\mathbf{A}\cdot\boldsymbol{\sigma})(\mathbf{B}\cdot\boldsymbol{\sigma}) &= \left(\sum_i A_i\sigma_i\right)\left(\sum_j B_j\sigma_j\right)\\ &= \sum_{i,j} A_iB_j\sigma_i\sigma_j\\ &= \sum_i A_iB_i\sigma_i^2 + \sum_{(i,j)\in\mathcal{C}} A_iB_j\sigma_i\sigma_j + \sum_{(i,j)\in\mathcal{C}} A_jB_i\sigma_j\sigma_i,\end{aligned}$$

where $\mathcal{C} = \{(1,2),(2,3),(3,1)\}$. The diagonal part is $(\mathbf{A}\cdot\mathbf{B})I$. Pairing the off-diagonal terms,

$$\begin{aligned}\sum_{i\neq j} A_iB_j\sigma_i\sigma_j &= \sum_{(i,j)\in\mathcal{C}} (A_iB_j\sigma_i\sigma_j + A_jB_i\sigma_j\sigma_i)\\ &= \sum_{(i,j)\in\mathcal{C}} (A_iB_j - A_jB_i)\sigma_i\sigma_j\\ &= \mathrm{i}\big[(A_2B_3 - A_3B_2)\sigma_1 + (A_3B_1 - A_1B_3)\sigma_2\\ &\qquad + (A_1B_2 - A_2B_1)\sigma_3\big]\\ &= \mathrm{i}(\mathbf{A}\times\mathbf{B})\cdot\boldsymbol{\sigma}.\end{aligned}$$

Orbital magnetic moment in quantum theory

A particle of mass m and charge q in a magnetic field $\mathbf{B}(\mathbf{R}) = \boldsymbol{\nabla}_R \times \mathbf{A}$ has, in Gaussian units,

$$\begin{aligned} H &= \frac{(\mathbf{P} - q\mathbf{A}/c)^2}{2m} \\ &= \underbrace{\frac{\mathbf{P}^2}{2m}}_{\text{zeroth order}} \underbrace{- \frac{q}{2mc}(\mathbf{P}\cdot\mathbf{A} + \mathbf{A}\cdot\mathbf{P})}_{\text{first order in } \mathbf{A}/c} + \underbrace{\frac{q^2\mathbf{A}^2}{2mc^2}}_{\text{second order}} . \end{aligned}$$

Choose the Coulomb gauge, $\boldsymbol{\nabla}\cdot\mathbf{A} = 0$. With $\langle \mathbf{r} \mid \psi\rangle = \psi(\mathbf{r})$,

$$\begin{aligned} \langle \mathbf{r}|\mathbf{P}\cdot\mathbf{A}|\psi\rangle &= -\mathrm{i}\hbar\boldsymbol{\nabla}\cdot\big(\mathbf{A}(\mathbf{r})\psi(\mathbf{r})\big) \\ &= -\mathrm{i}\hbar\big[(\boldsymbol{\nabla}\cdot\mathbf{A})\psi + \mathbf{A}\cdot\boldsymbol{\nabla}\psi\big] \\ &= \mathbf{A}(\mathbf{r})\cdot(-\mathrm{i}\hbar\boldsymbol{\nabla}\psi) = \langle \mathbf{r}|\mathbf{A}\cdot\mathbf{P}|\psi\rangle. \end{aligned}$$

Thus $\mathbf{P}\cdot\mathbf{A} = \mathbf{A}\cdot\mathbf{P}$ in this gauge. For a uniform field $\mathbf{B} = B\hat{\mathbf{e}}_z$, choose

$$\mathbf{A} = \frac{B}{2}(-Y\hat{\mathbf{e}}_x + X\hat{\mathbf{e}}_y), \quad \boldsymbol{\nabla}\cdot\mathbf{A} = 0, \quad \boldsymbol{\nabla}\times\mathbf{A} = B\hat{\mathbf{e}}_z.$$

The term linear in the field becomes

$$\begin{aligned} H_1 &= -\frac{q}{mc}\mathbf{A}\cdot\mathbf{P} = -\frac{q}{mc}\frac{B}{2}(-YP_x + XP_y) \\ &= -\frac{q}{2mc}\mathbf{L}\cdot\mathbf{B} = -\boldsymbol{\mu}\cdot\mathbf{B}, \qquad \boldsymbol{\mu} = \frac{q}{2mc}\mathbf{L}. \end{aligned}$$

In an L_z eigenstate,

$$\mu_z = \frac{q}{2mc}L_z, \qquad \mu_z \text{ has eigenvalues } \frac{q\hbar}{2mc}(0, \pm 1, \pm 2, \ldots).$$

The corresponding positive magneton scales are

$$\mu_B = \frac{e\hbar}{2m_e c}, \qquad \mu_N = \frac{e\hbar}{2m_p c}.$$

Clarification. The standard nuclear magneton uses the proton mass m_p. Replacing it by a general nucleon mass M, as in the source's mnemonic, defines the analogous scale $e\hbar/(2Mc)$ rather than the conventional definition of μ_N. Here $e > 0$; an electron has charge $q = -e$.

Spin magnetic moment and precession

Experiments show that electron spin also carries a magnetic moment:

$$\boldsymbol{\mu} = g\left(-\frac{e}{2mc}\right)\mathbf{S}, \qquad H_1 = -\boldsymbol{\mu}\cdot\mathbf{B} = \frac{ge}{2mc}\mathbf{S}\cdot\mathbf{B} = \frac{ge\hbar}{4mc}\boldsymbol{\sigma}\cdot\mathbf{B}.$$

The source quotes $g \simeq 2 \times 1.001159652 \simeq 2$. Define the signed gyromagnetic ratio $\gamma = -ge/(2mc)$; for $g \simeq 2$, $\gamma \simeq -e/(mc)$ and $H_1 = -\gamma\mathbf{S}\cdot\mathbf{B}$.

Time evolution of the spin state

For a constant magnetic field,

$$|\chi(t)\rangle = U(t)|\chi(0)\rangle, \qquad U(t) = \mathrm{e}^{-\mathrm{i}H_1 t/\hbar} = \mathrm{e}^{\mathrm{i}\gamma t\mathbf{S}\cdot\mathbf{B}/\hbar}.$$

Set $\mathbf{B} = B\hat{\mathbf{e}}_z$ and $\omega_0 = \gamma B$. In the S_z representation,

$$\begin{aligned} U(t) &= \begin{pmatrix} \langle +|\mathrm{e}^{\mathrm{i}\omega_0 t S_z/\hbar}|+\rangle & \langle +|\mathrm{e}^{\mathrm{i}\omega_0 t S_z/\hbar}|-\rangle \\ \langle -|\mathrm{e}^{\mathrm{i}\omega_0 t S_z/\hbar}|+\rangle & \langle -|\mathrm{e}^{\mathrm{i}\omega_0 t S_z/\hbar}|-\rangle \end{pmatrix} \\ &= \begin{pmatrix} \mathrm{e}^{\mathrm{i}\omega_0 t/2} & 0 \\ 0 & \mathrm{e}^{-\mathrm{i}\omega_0 t/2} \end{pmatrix}. \end{aligned}$$

For the initial state pointing along (θ, ϕ),

$$|\chi(0)\rangle = |\hat{\mathbf{n}}, +\rangle = \begin{pmatrix} \cos(\theta/2)\mathrm{e}^{-\mathrm{i}\phi/2} \\ \sin(\theta/2)\mathrm{e}^{\mathrm{i}\phi/2} \end{pmatrix},$$

and therefore

$$|\chi(t)\rangle = \begin{pmatrix} \cos(\theta/2)\mathrm{e}^{-\mathrm{i}(\phi-\omega_0 t)/2} \\ \sin(\theta/2)\mathrm{e}^{\mathrm{i}(\phi-\omega_0 t)/2} \end{pmatrix}.$$

The polar angle remains fixed, while the azimuth changes from ϕ to $\phi - \omega_0 t$. The sign follows the signed definition of γ above.

Time-independent perturbation theory

Suppose

$$H = H^0 + H', \qquad H^0|n^0\rangle = E_n^0|n^0\rangle, \qquad H|n\rangle = E_n|n\rangle.$$

The unperturbed problem is solvable, while H' is small compared with the relevant scales of H^0. Expand the energy and state in orders of the perturbation:

$$|n\rangle = |n^0\rangle + |n^1\rangle + |n^2\rangle + \cdots = \sum_{k=0}^{\infty} |n^k\rangle,$$
$$E_n = E_n^0 + E_n^1 + E_n^2 + \cdots = \sum_{k=0}^{\infty} E_n^k.$$

A superscript k denotes a kth-order correction, not a power. We hope successive terms become smaller in the regime of interest.

Nondegenerate starting state

Substitute the expansions into the exact eigenvalue equation:

$$(H^0 + H')[|n^0\rangle + |n^1\rangle + \cdots] = (E_n^0 + E_n^1 + \cdots)[|n^0\rangle + |n^1\rangle + \cdots].$$

Collecting equal orders gives

$$\begin{aligned} \text{zeroth order:} \quad & H^0|n^0\rangle = E_n^0|n^0\rangle, \\ \text{first order:} \quad & H^0|n^1\rangle + H'|n^0\rangle = E_n^0|n^1\rangle + E_n^1|n^0\rangle. \end{aligned}$$

Multiply the first-order equation by $\langle n^0|$. Since $\langle n^0|H^0 = E_n^0\langle n^0|$, the terms containing $|n^1\rangle$ cancel:

$$\boxed{E_n^1 = \langle n^0|H'|n^0\rangle.}$$

The first-order state correction

Project the first-order equation onto $\langle m^0|$, where $m \neq n$:

$$\langle m^0|H^0|n^1\rangle + \langle m^0|H'|n^0\rangle = E_n^0\langle m^0 \mid n^1\rangle.$$

Using $\langle m^0|H^0 = E_m^0\langle m^0|$ gives

$$\langle m^0 \mid n^1\rangle = \frac{\langle m^0|H'|n^0\rangle}{E_n^0 - E_m^0}.$$

The expansion of the correction thus has the form

$$|n^1\rangle = \sum_m |m^0\rangle\langle m^0 \mid n^1\rangle = \underbrace{\sum_{m\neq n} \frac{|m^0\rangle\langle m^0|H'|n^0\rangle}{E_n^0 - E_m^0}}_{|n_\perp^1\rangle} + |n_\parallel^1\rangle,$$

where $|n_\parallel^1\rangle \parallel |n^0\rangle$ and $|n_\perp^1\rangle \perp |n^0\rangle$.

Clarification. The perpendicular relation includes the explicit correction written on the source page.

Normalization, to first order, requires

$$\begin{aligned} 1 = \langle n \mid n\rangle &= \big(\langle n^0| + \langle n_\parallel^1| + \langle n_\perp^1|\big)\big(|n^0\rangle + |n_\parallel^1\rangle + |n_\perp^1\rangle\big) + O((H')^2) \\ &= 1 + \langle n_\parallel^1 \mid n^0\rangle + \langle n^0 \mid n_\parallel^1\rangle + O((H')^2). \end{aligned}$$

Hence $\langle n^0 \mid n_\parallel^1\rangle = \mathrm{i}\alpha$, with α real and first order in H'. Since $1 + \mathrm{i}\alpha = \mathrm{e}^{\mathrm{i}\alpha} + O(\alpha^2)$,

$$|n\rangle = \mathrm{e}^{\mathrm{i}\alpha}|n^0\rangle + \sum_{m\neq n} \frac{|m^0\rangle\langle m^0|H'|n^0\rangle}{E_n^0 - E_m^0} + O((H')^2).$$

Redefining the overall phase, $|n\rangle \mapsto \mathrm{e}^{-\mathrm{i}\alpha}|n\rangle$, therefore yields

$$\boxed{|n\rangle = |n^0\rangle + \sum_{m\neq n} \frac{|m^0\rangle\langle m^0|H'|n^0\rangle}{E_n^0 - E_m^0} + O((H')^2).}$$

Second-order energy and the validity criterion

The second-order energy correction is

$$\begin{aligned} E_n^2 &= \langle n^0|H'|n^1\rangle \\ &= \sum_{m\neq n} \frac{\langle n^0|H'|m^0\rangle\langle m^0|H'|n^0\rangle}{E_n^0 - E_m^0} \\ &= \boxed{\sum_{m\neq n} \frac{\left|\langle n^0|H'|m^0\rangle\right|^2}{E_n^0 - E_m^0}}, \end{aligned}$$

where Hermiticity of H' is used in the last step.

When is the expansion useful?

A necessary requirement is that the first-order state correction be small compared with the unperturbed state. In particular, each mixing coefficient should satisfy

$$\left|\frac{\langle m^0|H'|n^0\rangle}{E_n^0 - E_m^0}\right| \ll 1 \qquad (m \neq n).$$

This depends on three things:

1. The overall magnitude of H'.
2. The matrix element $\langle m^0|H'|n^0\rangle$, which also depends on the nature of the two states.
3. The energy spacing $|E_n^0 - E_m^0|$.

If $E_n^0 = E_m^0$, the nondegenerate expression has a zero denominator. One must use degenerate perturbation theory instead.

Clarification. Small individual coefficients are a useful necessary check; the norm of their combined correction must also remain small.

Exercise retained from the notes

Consider a charged harmonic oscillator in a uniform electric field $\mathcal{E}$:

$$H = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2 - q\mathcal{E}X.$$

Solve the problem both by first-order perturbation theory and exactly. The handwritten reading reminder refers to textbook pages 454–456; the textbook and edition are not identified on this page.

Selection rules

Parity

For a rotationally invariant H^0, label states by $|\alpha\ell m\rangle$. Their angular dependence is

$$\langle \mathbf{r} \mid \alpha\ell m\rangle \propto Y_\ell^m(\theta,\phi), \qquad Y_\ell^m(\pi-\theta,\pi+\phi) = (-1)^\ell Y_\ell^m(\theta,\phi).$$

Thus parity acts according to

$$\langle \mathbf{r}|\Pi|\alpha\ell m\rangle = (-1)^\ell \langle \mathbf{r} \mid \alpha\ell m\rangle.$$

Decompose a perturbation into its even and odd parts,

$$H' = H'_0 + H'_1, \qquad \Pi H'_p \Pi^{-1} = (-1)^p H'_p, \qquad p = 0, 1.$$

Then

$$\langle \alpha_2\ell_2 m_2|H'_p|\alpha_1\ell_1 m_1\rangle = 0 \quad \text{if } p+\ell_1+\ell_2 \text{ is odd.}$$

If the sum is even, parity alone does not determine whether the matrix element vanishes.

Angular momentum and spherical tensors

A perturbation may also be expanded into spherical-tensor components T_L^M. A rank-L tensor component carries the angular-momentum labels (L, M). For a multiplicative angular tensor, its coordinate action has the form

$$\langle \theta,\phi|T_L^M|\psi\rangle \propto Y_L^M(\theta,\phi)\langle \theta,\phi \mid \psi\rangle.$$

The angular-momentum selection rules following from the Wigner–Eckart theorem are

$$\langle \alpha_2\ell_2 m_2|T_L^M|\alpha_1\ell_1 m_1\rangle = 0 \quad \text{unless} \quad \begin{cases} |\ell_1 - L| \le \ell_2 \le \ell_1 + L, \\ m_2 = m_1 + M. \end{cases}$$

Clarification. The coordinate multiplication formula describes a multiplicative tensor, such as an angular factor in a potential; it is not the definition of every spherical-tensor operator. The parity label p and the tensor rank L are kept distinct here.

Hydrogen in a uniform electric field

Let the external electric potential be

$$\varphi_{\text{ext}}(\mathbf{r}) = -\mathbf{E}\cdot\mathbf{r}, \qquad -\boldsymbol{\nabla}\varphi_{\text{ext}} = \mathbf{E}.$$

If the electron is at $\mathbf{r}_1$ and the proton at $\mathbf{r}_2$, the perturbation is

$$\begin{aligned} H' &= -e\varphi_{\text{ext}}(\mathbf{r}_1) + e\varphi_{\text{ext}}(\mathbf{r}_2) \\ &= e\big[\varphi_{\text{ext}}(\mathbf{r}_2) - \varphi_{\text{ext}}(\mathbf{r}_1)\big] \\ &= e(\mathbf{r}_1 - \mathbf{r}_2)\cdot\mathbf{E} = e\mathbf{r}\cdot\mathbf{E}, \qquad \mathbf{r} \equiv \mathbf{r}_1 - \mathbf{r}_2. \end{aligned}$$

In a state of definite orbital parity,

$$\langle n\ell m|e\mathbf{r}\cdot\mathbf{E}|n\ell m\rangle = e\mathbf{E}\cdot\langle n\ell m|\mathbf{r}|n\ell m\rangle = 0.$$

Indeed, $\mathbf{r}$ is an odd-parity operator, so $1+\ell+\ell$ is odd. It is also a rank-one tensor, with components proportional to $T_1^{m'}$.

Clarification. A zero diagonal matrix element does not rule out a linear Stark effect in an excited, degenerate hydrogen level. For $n \geq 2$, the perturbation must first be diagonalized within the degenerate subspace.

Ground-state second-order shift

For $\mathbf{E} = E_0\hat{\mathbf{e}}_z$, $H' = eE_0 Z$, and

$$E^2_{100} = \sum_{n\ell m\neq 100} \frac{e^2E_0^2\,|\langle n\ell m|Z|100\rangle|^2}{E^0_{100} - E^0_{n\ell m}}.$$

Since $Z \propto Y_1^0$ is the $M = 0$ component of a vector,

- parity requires ℓ to be odd;
- the triangle rule gives $|0-1| \leq \ell \leq 0+1$, hence $\ell = 1$;
- the magnetic selection rule gives $m = 0+0 = 0$.

For the bound-state terms, this reduces to

$$\sum_{n=2}^{\infty} \frac{e^2E_0^2\,|\langle n10|Z|100\rangle|^2}{-\text{Ry}\,(1-1/n^2)}.$$

Clarification. The bound-state sum displayed in the source is only part of the complete spectral sum: continuum states with the same angular selection rules also contribute. No continuum evaluation is supplied on this page.

Degenerate perturbation theory

When $E_n^0 = E_m^0$, the criterion

$$\left| \frac{\langle m^0 | H' | n^0 \rangle}{E_n^0 - E_m^0} \right| \ll 1$$

breaks down. To understand the choice of basis, introduce $H(\lambda) = H^0 + \lambda H'$ and increase λ from 0 to 1. At $\lambda = 0$ there is freedom to choose any orthonormal basis of a degenerate eigenspace. Not every such basis connects smoothly to the eigenstates selected when the perturbation is turned on.

$$\begin{aligned} \lambda = 0: \quad & |\bar{n}^0\rangle, |\bar{m}^0\rangle, \quad E_n^0 = E_m^0 \\ & \Big\downarrow \text{ suitable basis: smooth branches} \\ \lambda = 1: \quad & |n\rangle, |m\rangle, \quad E_n \neq E_m \end{aligned}$$

A two-state example

Let

$$H^0 = \begin{pmatrix} E & 0 \\ 0 & E \end{pmatrix}, \qquad H' = \begin{pmatrix} 0 & \Delta \\ \Delta & 0 \end{pmatrix}, \qquad \Delta \in \mathbb{R}.$$

The exact eigenvalues follow from

$$\det \begin{pmatrix} E - \varepsilon & \Delta \\ \Delta & E - \varepsilon \end{pmatrix} = 0, \qquad \varepsilon = E \pm \Delta,$$

with eigenstates

$$|\pm\rangle = \frac{1}{\sqrt{2}} (|1^0\rangle \pm |2^0\rangle).$$

Choose instead the unperturbed basis

$$|\bar{1}^0\rangle = \frac{|1^0\rangle - |2^0\rangle}{\sqrt{2}}, \qquad |\bar{2}^0\rangle = \frac{|1^0\rangle + |2^0\rangle}{\sqrt{2}}.$$

In this ordered basis,

$$H' = \begin{pmatrix} -\Delta & 0 \\ 0 & \Delta \end{pmatrix}.$$

The perturbed eigenvectors are the same symmetric and antisymmetric combinations, now already selected by the unperturbed basis.

Clarification. The source writes these basis vectors in the order antisymmetric, symmetric, but reverses their eigenvalues in its drawn diagonal matrix. The ordering is made consistent above.

Choosing a basis in the degenerate subspace

Let D be the eigenspace of H^0 with energy E_n^0. Choose an orthonormal basis $\{|\bar{n}^0\rangle\}$ that diagonalizes H' within D:

$$\langle \bar{m}^0|H'|\bar{n}^0\rangle = \langle \bar{n}^0|H'|\bar{n}^0\rangle\,\delta_{mn} \quad \text{for } E_m^0 = E_n^0.$$

The first-order equation is still

$$H^0|n^1\rangle + H'|\bar{n}^0\rangle = E_n^0|n^1\rangle + E_n^1|\bar{n}^0\rangle.$$

For $E_m^0 \neq E_n^0$, projection onto $\langle m^0|$ yields

$$E_m^0\langle m^0 \mid n^1\rangle + \langle m^0|H'|\bar{n}^0\rangle = E_n^0\langle m^0 \mid n^1\rangle,$$
$$\langle m^0 \mid n^1\rangle = \frac{\langle m^0|H'|\bar{n}^0\rangle}{E_n^0 - E_m^0}.$$

Thus

$$|n^1\rangle = \underbrace{\sum_{E_m^0 \neq E_n^0} \frac{|m^0\rangle\langle m^0|H'|\bar{n}^0\rangle}{E_n^0 - E_m^0}}_{|n_\perp^1\rangle \in D^\perp} + \underbrace{|n_\parallel^1\rangle}_{\in D}.$$

For a different basis vector in D, the projected equation instead reads

$$(E_n^0 - E_m^0)\langle \bar{m}^0 \mid n^1\rangle = \langle \bar{m}^0|H'|\bar{n}^0\rangle, \qquad m \neq n,$$

which reduces to $0 = 0$ after diagonalizing H' in D. This includes the zero added as an annotation on the source page.

Normalization fixes $\mathrm{Re}\langle \bar{n}^0 \mid n_\parallel^1\rangle = 0$. Its component along $|\bar{n}^0\rangle$ is therefore $\mathrm{i}\alpha|\bar{n}^0\rangle$, with real α, and can be removed by $|n\rangle \mapsto \mathrm{e}^{-\mathrm{i}\alpha}|n\rangle$. The part determined by the first-order equation is

$$|n\rangle = |\bar{n}^0\rangle + \sum_{E_m^0 \neq E_n^0} \frac{|m^0\rangle\langle m^0|H'|\bar{n}^0\rangle}{E_n^0 - E_m^0} + |\eta_D\rangle + O((H')^2),$$

where $|\eta_D\rangle \in D$ is orthogonal to $|\bar{n}^0\rangle$.

Clarification. The source drops the entire within-subspace correction using an overall-phase argument. Only the parallel component along the chosen state can be removed that way. Mixing with other vectors in D is not fixed by this first-order equation; it generally requires higher-order equations. The source's final sum correctly gives the correction outside the degenerate subspace.

Blank source page

The supplied note page 76 is a blank lined sheet. It is retained here as a page marker so that the original sequence remains traceable.

The linear Stark effect for hydrogen at n = 2

Ignoring spin, the hydrogen $n = 2$ subspace contains the four degenerate states $|200\rangle$, $|210\rangle$, $|211\rangle$, and $|21,-1\rangle$. Although every diagonal element

$$\langle 2\ell m|eE_0Z|2\ell m\rangle = 0$$

vanishes by parity, the off-diagonal matrix elements need not vanish. In the ordered basis

$$\{|200\rangle, |210\rangle, |211\rangle, |21,-1\rangle\},$$

the perturbation is

$$H'|_{n=2} = \begin{pmatrix} 0 & \Delta & 0 & 0 \\ \Delta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \Delta = \langle 200|eE_0Z|210\rangle.$$

The phases of the basis vectors may be chosen so that Δ is real.

Exercise retained from the notes

Calculate Δ explicitly using the hydrogen eigenfunctions. The source suggests its textbook's Eq. (13.1.27); the textbook and edition are not specified on this page.

Diagonalization

Use the basis

$$\left\{ \frac{|200\rangle - |210\rangle}{\sqrt{2}}, \quad \frac{|200\rangle + |210\rangle}{\sqrt{2}}, \quad |211\rangle, \quad |21,-1\rangle \right\}.$$

The perturbation becomes

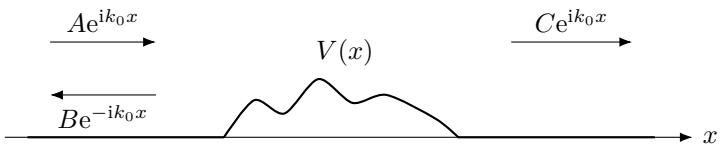
$$H'|_{n=2} = \mathrm{diag}(-\Delta, \Delta, 0, 0).$$

The corresponding first-order shifts are $-\Delta$, Δ, 0, and 0. In this basis, off-diagonal perturbation elements between different vectors of the original degenerate subspace vanish.

Clarification. As in note 74, the source's drawn diagonal matrix has the signs interchanged relative to its stated order of the symmetric and antisymmetric states. The matrix and basis order agree here.

One-dimensional scattering

A stationary scattering eigenfunction of $H = T+V$ has an incident wave, a reflected wave, and a transmitted wave. For a localized potential, the asymptotic wave number is k_0 on both sides.



The time-dependent picture uses wave packets:

1. Start with a normalized incident packet $|\Psi\rangle$ whose mean momentum is $\langle\Psi|P|\Psi\rangle = \hbar k_0$.
2. Expand it in scattering eigenstates:

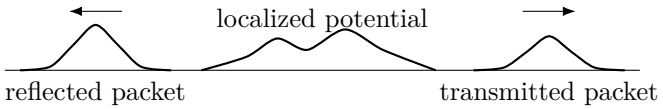
$$|\Psi\rangle = \int a(k)|\psi_k\rangle \, \mathrm{d}k, \qquad H|\psi_k\rangle = E_k|\psi_k\rangle.$$

3. Evolve each energy component:

$$|\Psi(t)\rangle = \int a(k)\mathrm{e}^{-\mathrm{i}E_k t/\hbar}|\psi_k\rangle \, \mathrm{d}k.$$

4. Long after scattering, the packet separates into reflected and transmitted parts:

$$|\Psi(t\to\infty)\rangle = |\Psi_R\rangle + |\Psi_T\rangle, \qquad R = \langle\Psi_R \mid \Psi_R\rangle = \int |\Psi_R(x)|^2 \, \mathrm{d}x.$$



Equivalently, one solves the stationary problem:

$$\psi(x) = \begin{cases} Ae^{ik_0x} + Be^{-ik_0x}, & \text{left of the interaction region,} \\ \text{solution determined by } V(x), & \text{inside the interaction region,} \\ Ce^{ik_0x}, & \text{right of the interaction region.} \end{cases} \qquad R = \frac{|B|^2}{|A|^2}.$$

Clarification. The stationary reflection coefficient is a ratio of probability currents, hence squared amplitudes. The displayed formula assumes the same asymptotic wave number on both sides.

Three-dimensional scattering

The differential cross section relates the rate of scattering into a small solid angle $\mathrm{d}\Omega$ to the incident particle flux:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\,\mathrm{d}\Omega = \mathrm{d}\sigma = \frac{\text{number scattered into } \mathrm{d}\Omega \text{ per unit time}}{\text{number incident per unit time per unit area}}.$$

To calculate it quantum mechanically, consider an incident plane wave $\mathrm{e}^{\mathrm{i}kz}$ and an eigenfunction of $H = T+V$ of the form

$$\psi = \mathrm{e}^{\mathrm{i}kz} + \psi_{\mathrm{sc}}(r,\theta,\phi).$$

The scattered wave is centered on the scattering center. Far outside the potential, both terms satisfy the free Helmholtz equation at the same energy:

$$(\nabla^2 + k^2)\mathrm{e}^{\mathrm{i}kz} = 0, \qquad (\nabla^2 + k^2)\psi_{\mathrm{sc}} = 0 \quad (r \to \infty).$$

The scattered wave can be expanded in spherical harmonics:

$$\psi_{\mathrm{sc}} \sim \sum_{\ell,m} [A_{\ell m} j_\ell(kr) + B_{\ell m} n_\ell(kr)] Y_\ell^m(\theta,\phi).$$

Here j_ℓ and n_ℓ are spherical Bessel and Neumann functions. Their large-r limits are

$$j_\ell(kr) \sim \frac{\sin(kr - \ell\pi/2)}{kr}, \qquad n_\ell(kr) \sim -\frac{\cos(kr - \ell\pi/2)}{kr}.$$

Consequently, each radial combination has the asymptotic form

$$A_{\ell m} j_\ell(kr) + B_{\ell m} n_\ell(kr) \sim C_{\ell m}\frac{\mathrm{e}^{\mathrm{i}kr}}{r} + D_{\ell m}\frac{\mathrm{e}^{-\mathrm{i}kr}}{r}.$$

Including the time factor gives

$$C_{\ell m}\frac{\mathrm{e}^{\mathrm{i}kr - \mathrm{i}\omega t}}{r} + D_{\ell m}\frac{\mathrm{e}^{-\mathrm{i}kr - \mathrm{i}\omega t}}{r}.$$

The first term is outgoing, while the second is incoming. The physical scattering setup chooses an outgoing scattered wave.

Clarification. The radial expansion coefficients are written with both ℓ and m indices for a general angular distribution. The source abbreviates these coefficients by a single ℓ index. Its reading reminder gives textbook page 348 without identifying an edition.

Scattering amplitude and differential cross section

Keeping only the outgoing radial wave gives

$$\begin{aligned}\psi_{\mathrm{sc}} &\sim \frac{\mathrm{e}^{\mathrm{i}kr}}{kr}\sum_{\ell,m}(-\mathrm{i})^{\ell}(-B_{\ell m})Y_{\ell}^{m}(\theta,\phi)\\ &= \frac{\mathrm{e}^{\mathrm{i}kr}}{r}f(\theta,\phi),\end{aligned}$$

which defines the scattering amplitude f. Thus

$$\boxed{\psi_k \sim \mathrm{e}^{\mathrm{i}kz} + f(\theta,\phi)\frac{\mathrm{e}^{\mathrm{i}kr}}{r}.}$$

For a finite beam of transverse extent $\rho_{\max}$, the notes distinguish the forward incident-beam cone, $\theta \lesssim \rho_{\max}/r$, from detectors outside that cone. The incident plane-wave current is

$$\mathbf{j}_{\mathrm{inc}} = \frac{\hbar}{2\mu\mathrm{i}}\left(\mathrm{e}^{-\mathrm{i}kz}\boldsymbol{\nabla}\mathrm{e}^{\mathrm{i}kz} - \mathrm{e}^{\mathrm{i}kz}\boldsymbol{\nabla}\mathrm{e}^{-\mathrm{i}kz}\right) = \frac{\hbar k}{\mu}\hat{\mathbf{e}}_z,$$

where μ is the reduced mass. For the scattered wave,

$$\mathbf{j}_{\mathrm{sc}} = \frac{\hbar}{2\mu\mathrm{i}}(\psi_{\mathrm{sc}}^{*}\boldsymbol{\nabla}\psi_{\mathrm{sc}} - \psi_{\mathrm{sc}}\boldsymbol{\nabla}\psi_{\mathrm{sc}}^{*}).$$

Using the spherical-coordinate gradient,

$$\boldsymbol{\nabla} = \hat{\mathbf{r}}\frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial}{\partial\theta} + \hat{\boldsymbol{\phi}}\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi},$$

we find

$$\boldsymbol{\nabla}\left(f\frac{\mathrm{e}^{\mathrm{i}kr}}{r}\right) = \hat{\mathbf{r}}\,\mathrm{i}kf\frac{\mathrm{e}^{\mathrm{i}kr}}{r} + O(r^{-2}), \qquad \mathbf{j}_{\mathrm{sc}} \sim \hat{\mathbf{r}}\,\frac{|f|^2}{r^2}\frac{\hbar k}{\mu}.$$

The probability flow into $\mathrm{d}\Omega$ per unit time is therefore

$$\mathcal{R}(\theta,\phi;\mathrm{d}\Omega) = \mathbf{j}_{\mathrm{sc}}\cdot\hat{\mathbf{r}}\,r^2\,\mathrm{d}\Omega = |f|^2\frac{\hbar k}{\mu}\,\mathrm{d}\Omega.$$

Dividing by the incident flux yields

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\,\mathrm{d}\Omega = \frac{\mathcal{R}}{j_{\mathrm{inc}}} = |f(\theta,\phi)|^2\,\mathrm{d}\Omega, \qquad \boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = |f(\theta,\phi)|^2.}$$

Introduction to Green's functions

Source reading reminder: Sec. 8.2, pp. 121–123; the textbook is not identified.

For a time-independent one-particle problem, let

$$H_0(\mathbf{r}) = -\frac{\hbar^2}{2m}\nabla^2 + V_0(\mathbf{r}), \qquad [E - H_0(\mathbf{r})]\psi_E^0(\mathbf{r}) = 0.$$

Define a Green's function by

$$[E - H_0(\mathbf{r})]G_0(\mathbf{r}, \mathbf{r}'; E) = \delta^{(3)}(\mathbf{r} - \mathbf{r}').$$

We wish to solve

$$[E - H_0(\mathbf{r}) - V(\mathbf{r})]\psi_E(\mathbf{r}) = 0.$$

Insert the delta-function identity into $V\psi_E$:

$$\begin{aligned} V(\mathbf{r})\psi_E(\mathbf{r}) &= \int \mathrm{d}^3r'\, \delta^{(3)}(\mathbf{r} - \mathbf{r}')V(\mathbf{r}')\psi_E(\mathbf{r}') \\ &= \int \mathrm{d}^3r'\, [E - H_0(\mathbf{r})]G_0(\mathbf{r}, \mathbf{r}'; E)V(\mathbf{r}')\psi_E(\mathbf{r}') \\ &= [E - H_0(\mathbf{r})] \int \mathrm{d}^3r'\, G_0(\mathbf{r}, \mathbf{r}'; E)V(\mathbf{r}')\psi_E(\mathbf{r}'). \end{aligned}$$

Since $V\psi_E = (E - H_0)\psi_E$, the difference is a homogeneous solution:

$$\psi_E(\mathbf{r}) = f(\mathbf{r}) + \int \mathrm{d}^3r'\, G_0(\mathbf{r}, \mathbf{r}'; E)V(\mathbf{r}')\psi_E(\mathbf{r}'), \qquad (E - H_0)f = 0.$$

For the prescribed incident channel choose $f = c\psi_E^0$. Requiring the same incident normalization when $V \to 0$ fixes $c = 1$:

$$\boxed{\psi_E = \psi_E^0 + G_0 V \psi_E.}$$

Iterating this integral equation gives the formal Born series,

$$\begin{aligned} \psi_E &= \psi_E^0 + G_0V(\psi_E^0 + G_0V\psi_E) \\ &= \psi_E^0 + G_0V\psi_E^0 + G_0VG_0V\psi_E^0 + \cdots \\ &= \sum_{n=0}^{\infty} (G_0V)^n \psi_E^0. \end{aligned}$$

Clarification. Boundary conditions select the homogeneous solution and the Green's function. A general homogeneous solution need not be a single multiple of one chosen ψ_E^0. The series is formal unless its convergence or a controlled truncation has been established.

Time-dependent Green's functions

This page uses units $\hbar = 1$. The free and full wave equations are

$$(\mathrm{i}\partial_t - H_0)\psi^0(\mathbf{r}, t) = 0, \qquad (\mathrm{i}\partial_t - H_0 - V)\psi(\mathbf{r}, t) = 0.$$

Their Green's functions satisfy

$$(\mathrm{i}\partial_t - H_0)G_0(\mathbf{r}t, \mathbf{r}'t') = \delta^{(3)}(\mathbf{r} - \mathbf{r}')\delta(t - t'),$$
$$(\mathrm{i}\partial_t - H_0 - V)G(\mathbf{r}t, \mathbf{r}'t') = \delta^{(3)}(\mathbf{r} - \mathbf{r}')\delta(t - t').$$

With matching boundary conditions, the two useful integral forms are

$$\psi(\mathbf{r}, t) = \psi^0(\mathbf{r}, t) + \int \mathrm{d}^3r'\,\mathrm{d}t'\,G_0(\mathbf{r}t, \mathbf{r}'t')V(\mathbf{r}')\psi(\mathbf{r}', t'),$$
$$\psi(\mathbf{r}, t) = \psi^0(\mathbf{r}, t) + \int \mathrm{d}^3r'\,\mathrm{d}t'\,G(\mathbf{r}t, \mathbf{r}'t')V(\mathbf{r}')\psi^0(\mathbf{r}', t').$$

Operator form and boundary conditions

For a time-independent full Hamiltonian H, put

$$U(\mathbf{r}t, \mathbf{r}'t') = \langle\mathbf{r}|\mathrm{e}^{-\mathrm{i}H(t-t')}|\mathbf{r}'\rangle.$$

The retarded and advanced functions are

$$G^R = -\mathrm{i}\Theta(t - t')U, \qquad G^A = +\mathrm{i}\Theta(t' - t)U.$$

Each satisfies the unit-delta equation, with a different time boundary condition. The piecewise expression written in the source is

$$K = G^R + G^A = \begin{cases} -\mathrm{i}U, & t > t', \\ +\mathrm{i}U, & t < t'. \end{cases}$$

Clarification. With the delta-function normalization stated above, $(\mathrm{i}\partial_t - H)K = 2\delta^{(3)}(\mathbf{r} - \mathbf{r}')\delta(t - t')$, not a unit delta. Thus the source's unrestricted claim that $\alpha G^R + \beta G^A$ is a Green's function needs the condition $\alpha + \beta = 1$. In particular, the normalized symmetric choice is $\frac{1}{2}(G^R + G^A)$. This follows directly from the jump at $t = t'$.

The Born approximation and a Fourier-space Green's function

For an incident plane wave, the first Born approximation is

$$\psi_{\mathbf{k}}(\mathbf{r}) = \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}} + \frac{2\mu}{\hbar^2}\int \mathrm{d}^3 r'\, G^{(0)}(\mathbf{r},\mathbf{r}')V(\mathbf{r}')\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}'} + O(V^2).$$

Here the Helmholtz Green's function is defined by

$$(\nabla^2 + k^2)G^{(0)}(\mathbf{r},\mathbf{r}') = \delta^{(3)}(\mathbf{r}-\mathbf{r}').$$

Translation invariance allows $G^{(0)}(\mathbf{r},\mathbf{r}') = G^{(0)}(\mathbf{x})$ with $\mathbf{x} = \mathbf{r}-\mathbf{r}'$:

$$(\nabla_x^2 + k^2)G^{(0)}(\mathbf{x}) = \delta^{(3)}(\mathbf{x}).$$

Fourier transform

Use the symmetric transform convention

$$\widetilde{G}^{(0)}(\mathbf{q}) = (2\pi)^{-3/2}\int \mathrm{d}^3 x\, \mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{x}}G^{(0)}(\mathbf{x}).$$

Transforming the differential equation and integrating the Laplacian by parts gives

$$\begin{aligned}(2\pi)^{-3/2}\int \mathrm{d}^3 x\, \mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{x}}(\nabla_x^2 + k^2)G^{(0)}(\mathbf{x}) &= (2\pi)^{-3/2},\\ (k^2 - q^2)\widetilde{G}^{(0)}(\mathbf{q}) &= (2\pi)^{-3/2}.\end{aligned}$$

The denominator vanishes on the shell $q^2 = k^2$. To select a boundary condition, use the prescription $k^2 - q^2 \mapsto k^2 + \mathrm{i}\epsilon - q^2$, with $\epsilon > 0$ taken to zero after integration:

$$\widetilde{G}^{(0)}(\mathbf{q}) = \frac{(2\pi)^{-3/2}}{k^2 + \mathrm{i}\epsilon - q^2}, \qquad G^{(0)}(\mathbf{x}) = (2\pi)^{-3/2}\int \mathrm{d}^3 q\, \mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{x}}\widetilde{G}^{(0)}(\mathbf{q}).$$

Clarification. For the free reference Hamiltonian, the energy-resolvent kernel of note 81 is $(2\mu/\hbar^2)G^{(0)}$. The notation here makes the factor in the Born integral explicit.

Radial Fourier integral and the contour prescription

Choose the polar axis in momentum space parallel to $\mathbf{x}$, and write $r = |\mathbf{x}| > 0$. Inserting the Fourier-space kernel,

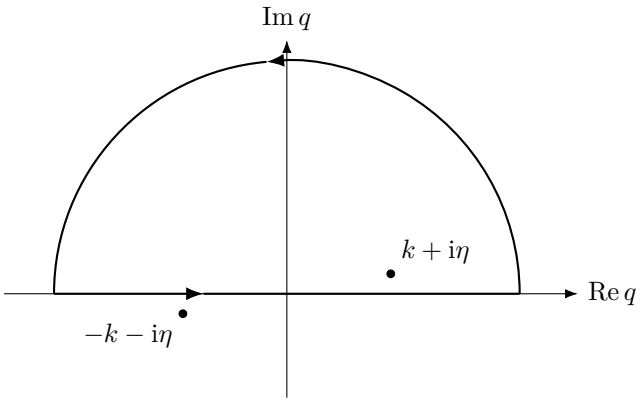
$$\begin{aligned} G^{(0)}(r) &= \frac{1}{8\pi^3} \int_0^\infty q^2 \, \mathrm{d}q \int_0^{2\pi} \mathrm{d}\phi \int_{-1}^{1} \mathrm{d}(\cos\theta) \, \frac{\mathrm{e}^{\mathrm{i}qr\cos\theta}}{k^2 + \mathrm{i}\epsilon - q^2} \\ &= \frac{1}{4\pi^2} \int_0^\infty \frac{\mathrm{e}^{\mathrm{i}qr} - \mathrm{e}^{-\mathrm{i}qr}}{\mathrm{i}qr} \frac{q^2 \, \mathrm{d}q}{k^2 + \mathrm{i}\epsilon - q^2} \\ &= \frac{1}{4\pi^2} \int_{-\infty}^\infty \frac{\mathrm{e}^{\mathrm{i}qr}}{\mathrm{i}qr} \frac{q^2 \, \mathrm{d}q}{k^2 + \mathrm{i}\epsilon - q^2} \\ &= -\frac{\mathrm{i}}{4\pi^2 r} \int_{-\infty}^\infty \frac{q \, \mathrm{e}^{\mathrm{i}qr}}{k^2 + \mathrm{i}\epsilon - q^2} \, \mathrm{d}q. \end{aligned}$$

Clarification. The handwritten last prefactor contains r^2 in its denominator. The preceding line gives a single factor r, as shown above.

Consider the integrand

$$\frac{q \, \mathrm{e}^{\mathrm{i}qr}}{k^2 + \mathrm{i}\epsilon - q^2}$$

in the complex q plane. For $\epsilon > 0$ and $k > 0$, its poles lie near $q = k + \mathrm{i}\eta$ and $q = -k - \mathrm{i}\eta$, where $\eta \simeq \epsilon/(2k) > 0$. Since $r > 0$, close the contour in the upper half-plane.



The supplied final page ends at this contour setup. It does not include the subsequent residue evaluation.